Problema 1 (segundo parcial)

domingo, 22 de diciembre de 2024 19:59

Calcula el flujo del campo vectorial $\overline{F}(y^2, xz^2, 0)$ hacia el interior de la superficie $S = S_1 \cup S_2$, donde $S_1 = \{(x,y,z) \in \mathbb{R}^3 \mid z^2 = x^2 + y^2,\ 0 \leqslant z \leqslant 2\}$ y $S_2 = \{(x,y,z) \in \mathbb{R}^3 \mid x^2 + y^2 \leqslant 4,\ z = 2\}$.

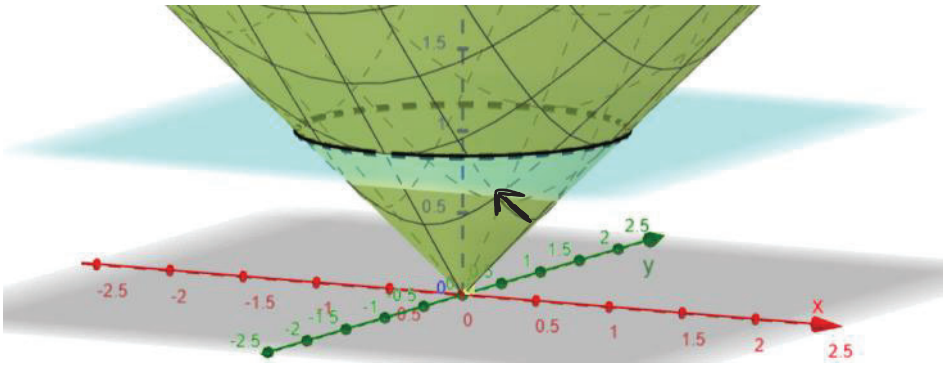
Parametrizamos las superficies:

$$S_1: \quad x^2+y^2 = z^2 \longrightarrow z = \pm\sqrt{x^2+y^2} \xrightarrow{z>0} z = \sqrt{x^2+y^2}$$

$$\bar{r}(t) = (u, v, \sqrt{u^2+v^2}) \Longrightarrow \begin{cases} \bar{r}_u(u,v) = \left(1, 0, \dfrac{u}{\sqrt{u^2+v^2}}\right) \\ \bar{r}_v(u,v) = \left(0, 1, \dfrac{v}{\sqrt{u^2+v^2}}\right) \end{cases}$$

$$\bar{r}_u \times \bar{r}_v = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ 1 & 0 & \dfrac{u}{\sqrt{u^2+v^2}} \\ 0 & 1 & \dfrac{v}{\sqrt{u^2+v^2}} \end{vmatrix} = \left(\frac{-u}{\sqrt{u^2+v^2}}, \frac{-v}{\sqrt{u^2+v^2}}, 1\right)$$

Vector "hacia arriba"



Que el vector normal se dirija "hacia arriba" implica en este caso que apunte al interior del cono. Debemos tomar el vector normal de la superficie S_2 apuntando también hacia el interior de la figura.

$$S_2: \quad z = 2 \longrightarrow \bar{r}(u,v) = (u\cdot\cos(v), u\cdot\sin(v), 2), \quad u \in [0,2],\ v \in [0,2\pi]$$

$$\bar{n} = (0,0,-1)$$

$$\text{Flujo} = \iint_S \bar{F}\cdot\bar{N}\,dS = \iint_{S_1} \bar{F}\cdot\bar{N}\cdot dS + \iint_{S_2} \bar{F}\cdot\bar{N}\cdot dS =$$

$$= \iint_{R_1} \left(v^2,\ u\cdot(u^2+v^2),\ 0\right)\cdot\left(\frac{-u}{\sqrt{u^2+v^2}},\ \frac{-v}{\sqrt{u^2+v^2}},\ 1\right)\cdot du\cdot dv +$$

$$+ \iint_{R_2} \left(u^2\cdot \operatorname{sen}^2(v),\ u\cdot\cos(v)\cdot 4,\ 0\right)\cdot(0,\ 0,\ -1)\cdot du\,dv =$$

$$0$$

$$= \iint_{R_1} \left(\frac{-uv^2}{\sqrt{u^2+v^2}} - \frac{u^3 v}{\sqrt{u^2+v^2}} - \frac{uv^3}{\sqrt{u^2+v^2}}\right) du\,dv = \iint_{R_1} \frac{-uv\left(v+u^2+v^2\right)}{\sqrt{u^2+v^2}}\,du\,dv =$$

$$= \int_{\theta=0}^{\theta=2\pi}\left(\int_{r=0}^{r=2} \frac{-r\cos(\theta)\, r\operatorname{sen}(\theta)\left(r\operatorname{sen}(\theta) + \overbrace{r^2\cos^2(\theta)+r^2\operatorname{sen}^2(\theta)}^{r^2}\right)}{r}\cdot dr\right) d\theta =$$

$$= -\int_{\theta=0}^{\theta=2\pi}\int_{r=0}^{r=2}\left(r^3\cdot\cos(\theta)\cdot\operatorname{sen}^2(\theta) + r^4\cdot\cos(\theta)\cdot\operatorname{sen}(\theta)\right) dr\, d\theta =$$

$$= -\int_{\theta=0}^{\theta=2\pi}\left[\frac{r^4}{4}\cos(\theta)\operatorname{sen}^2(\theta) + \frac{r^5}{5}\cos(\theta)\operatorname{sen}(\theta)\right]_0^2 d\theta =$$

$$= -\int_{\theta=0}^{\theta=2\pi}\left(4\cdot\cos(\theta)\operatorname{sen}^2(\theta) + \frac{32}{5}\cos(\theta)\operatorname{sen}(\theta)\right) d\theta =$$

$$= -\left[4\cdot\frac{\operatorname{sen}^3(\theta)}{3} + \frac{32}{5}\cdot\frac{\operatorname{sen}^2(\theta)}{2}\right]_0^{2\pi} = \boxed{0}$$

Problema 2 (segundo parcial)

domingo, 22 de diciembre de 2024 19:59

Calcula la integral de línea del campo vectorial $\overline{F}(x,y,z) = (3y^2z + ye^x, 6xyz + e^x, 3xy^2 + 1)$ a lo largo del segmento desde el punto $A = (1,1,1)$ hasta $B = (1,0,5)$.

Vamos a comprobar que el campo es conservativo:

① $P(x,y,z) = (3y^2z + ye^x,\ 6xyz + e^x,\ 3xy^2)$

$\frac{\partial P}{\partial y} = 6yz + e^x = \frac{\partial Q}{\partial x}$, $\frac{\partial P}{\partial z} = 3y^2 = \frac{\partial R}{\partial x}$, $\frac{\partial Q}{\partial z} = 6xy = \frac{\partial R}{\partial y}$

Luego $\overline{F}(x,y,z)$ es irrotacional en todo $\mathbb{R}^3$.

② El dominio de $\overline{F}(x,y,z)$, que es $\mathbb{R}^3$, es un conjunto simplemente conexo (alternativamente se puede justificar que $P(x,y,z)$, $Q(x,y,z)$, $R(x,y,z)$ son funciones de clase C^1 en el conjunto convexo $\mathbb{R}^3$).

Por lo tanto, $\overline{F}(x,y,z)$ es un campo conservativo en $\mathbb{R}^3$.

Calculamos ahora la función potencial sabiendo que $\overline{F}(x,y,z) = \nabla U(x,y,z)$

- $\frac{\partial U}{\partial x} = P(x,y,z) \Rightarrow U(x,y,z) = \int (3y^2z + ye^x)\,dx + C(y,z) = 3xy^2z + ye^x + C(y,z)$
- $\frac{\partial U}{\partial y} = Q(x,y,z) \Rightarrow 6xyz + e^x + \frac{\partial C(y,z)}{\partial y} = 6xyz + e^x \Rightarrow \frac{\partial C(y,z)}{\partial y} = 0 \Rightarrow C(y,z) = C(z)$
- $\frac{\partial U}{\partial z} = R(x,y,z) \Rightarrow 3xy^2 + C'(z) = 3xy^2 + 1 \Rightarrow C(z) = z + C$

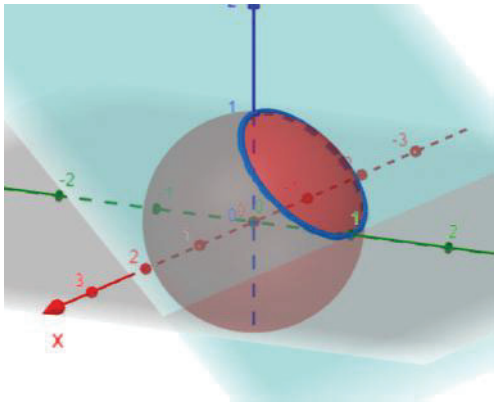
Luego $U(x,y,z) = 3xy^2z + ye^x + z + C$

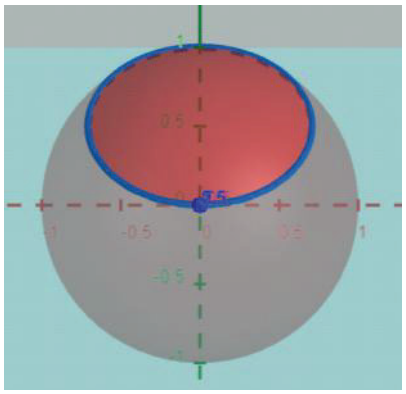
$$\int_C \overline{F}(x,y,z)\cdot d\overline{r} = U(B) - U(A) = 5 - (3 + e + 1) = \boxed{1 - e}$$

Problema 3 (segundo parcial)

domingo, 22 de diciembre de 2024 19:59

Calcula la integral $\int_C (ydx + zdy + xdz)$, siendo C la curva intersección de la esfera $x^2+y^2+z^2 = a^2$ con el plano $y + z = a$.





Calculamos primero la curva intersección:

$x^2+y^2+z^2 = a^2$, $y+z=a \rightarrow z = a-y$ $\longrightarrow$ $x^2+y^2+(a-y)^2=a^2 \longrightarrow x^2+y^2+a^2-2ay+y^2=a^2$

$\longrightarrow \quad x^2+2(y^2-ay)=0 \longrightarrow x^2+2\left(y-\frac{a}{2}\right)^2 = \frac{a^2}{2}$

Teorema de Stokes: $\oint_C \bar{F}(x,y,z)\cdot d\bar{r} = \iint_S (\nabla \times \bar{F})\cdot \bar{N}\cdot dS$

$$\bar{F}(x,y,z) = (y,z,x) \Longrightarrow \nabla \times \bar{F} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ y & z & x \end{vmatrix} = (-1,-1,-1)$$

Por otra parte, el plano que contiene la circunferencia limitada por C es $y+z=a \Rightarrow \bar{n} = (0,1,1) \Rightarrow \bar{N} = \left(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

Vamos a utilizar $z = f(x,y) = a-y$, con $R = \{(x,y)\in\mathbb{R}^2 \mid x^2+2\left(y-\frac{a}{2}\right)^2 \le \frac{a^2}{2}\}$, es decir, el recinto limitado por $\left(\frac{x}{a/\sqrt{2}}\right)^2 + \left(\frac{y-a/2}{a/2}\right)^2 = 1$.

$$\iint_S (\nabla\times\bar{F})\cdot\bar{N}\cdot dS = \iint_S (-1,-1,-1)\cdot\left(0,\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}\right)dS = -\frac{2}{\sqrt{2}}\iint_S dS =$$

$$= -\sqrt{2}\iint_R \sqrt{1+0^2+(-1)^2}\,dx\,dy = -2\iint_R dx\,dy = (*)$$

$$\left.\begin{array}{l} x = \frac{a}{\sqrt{2}} r\cos(\theta) \\ y = \frac{a}{2} + \frac{a}{2} r\,\mathrm{sen}(\theta) \end{array}\right] \Rightarrow |J| = \begin{vmatrix} \frac{a}{\sqrt{2}}\cos(\theta) & -\frac{a}{\sqrt{2}} r\,\mathrm{sen}(\theta) \\ \frac{a}{2}\,\mathrm{sen}(\theta) & \frac{a}{2} r\cos(\theta) \end{vmatrix} = \frac{a^2}{2\sqrt{2}} r$$

$$= (*) \quad -2\int_{\theta=0}^{\theta=2\pi}\left(\int_{r=0}^{r=1} \frac{a^2}{2\sqrt{2}} r\,dr\right) d\theta = -\cancel{2}\cdot\frac{a^2}{\cancel{2}\sqrt{2}}\left[\theta\right]_0^{2\pi}\left[\frac{r^2}{2}\right]_0^1 = \boxed{-\frac{\pi\cdot a^2}{\sqrt{2}}}$$

Nota: Alternativamente, se puede calcular la integral de línea de la curva $\alpha(t) = \left(\underbrace{\frac{a}{\sqrt{2}}\cos(t)}_{x(t)}, \underbrace{\frac{a}{2} + \frac{a}{2}\,\mathrm{sen}(t)}_{y(t)}, \underbrace{\frac{a}{2} - \frac{a}{2}\,\mathrm{sen}(t)}_{z(t)}\right)$

$$\oint_C (y\,dx + z\,dy + x\,dz) =$$

$$= \int_0^{2\pi}\left(\left(\frac{a}{2}+\frac{a}{2}\,\mathrm{sen}(t)\right)\cdot\left(-\frac{a}{\sqrt{2}}\,\mathrm{sen}(t)\right) + \left(\frac{a}{2}-\frac{a}{2}\,\mathrm{sen}(t)\right)\cdot\left(\frac{a}{2}\cos(t)\right) + \left(\frac{a}{\sqrt{2}}\cos(t)\right)\cdot\left(-\frac{a}{2}\cos(t)\right)\right) dt =$$

$$= \int_0^{2\pi}\left(-\frac{a^2}{2\sqrt{2}}\,\mathrm{sen}(t) - \frac{a^2}{2\sqrt{2}}\,\mathrm{sen}^2(t) + \frac{a^2}{4}\cos(t) - \frac{a^2}{4}\,\mathrm{sen}(t)\cos(t) - \frac{a^2}{2\sqrt{2}}\cos^2(t)\right) dt =$$

$$= -\frac{a^2}{2\sqrt{2}}\int_0^{2\pi}\cancel{\mathrm{sen}(t)} - \frac{a^2}{2\sqrt{2}}\int_0^{2\pi}\left(\frac{1-\cos(2t)}{2}\right) dt + \frac{a^2}{4}\int_0^{2\pi}\cancel{\cos(t)} - \frac{a^2}{4}\int_0^{2\pi}\cancel{\frac{\mathrm{sen}(2t)}{2}}\,dt - \frac{a^2}{2\sqrt{2}}\int_0^{2\pi}\left(\frac{1+\cos(2t)}{2}\right) dt =$$

$$= \frac{-a^2}{2\sqrt{2}}\int_0^{2\pi}\left(\frac{1}{2} - \cancel{\frac{\cos(2t)}{2}} + \frac{1}{2} + \cancel{\frac{\cos(2t)}{2}}\right) dt = \frac{-a^2}{2\sqrt{2}}\cdot\left[t\right]_0^{2\pi} = \boxed{-\frac{\pi\cdot a^2}{\sqrt{2}}}$$